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Polymer Degradation and Stability 91 (2006) 1989–1995
Polymer Degradation and Stability
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Kinetic analysis for hydrothermal depolymerization of nylon 6

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Received 10 November 2005; received in revised form 16 February 2006; accepted 21 February 2006
Available online 6 May 2006

Abstract

We investigated the influences of reaction temperature, time and water density. We also investigated the reaction rate and reaction scheme which is necessary to industrialize this process from the result of hydrolysis of nylon 6 in sub- and supercritical water. The reactions were carried out at temperatures between 573 and 673 K under the estimated pressures of up to 35 MPa (water amount charged in a reactor 0.83–3.79 g/5 cm^3 – reactor) for reaction times of 5–60 min. The main products of hydrolysis of nylon 6 by sub- and supercritical water are ε-caprolactam and ε-aminocaproic acid. Based on our experimental results, the reaction scheme of nylon 6 decomposition is represented. Nylon 6 was decomposed into ε-aminocaproic acid by hydrolysis followed by cyclodehydration to ε-caprolactam or decomposition further to lower molecules. At each reaction conditions, the k_1 and k_2 values were determined and the activation energy was calculated.
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Keywords: Depolymerization; Nylon 6; Supercritical water; Reaction mechanism

1. Introduction

Plastics have been used in various situations and its use is increasing year by year. Since waste plastics recycling has been seriously desired, the recycling technology has not well developed yet. Currently, landfilling and incineration are extensively used for waste plastics. Only a fraction of waste plastics has been recycled by material recycling or thermal energy recovery. Therefore, new technology for waste plastics recycling has been expected to develop because conventional methods have problems.

Recycling methods of waste plastics are classified into material recycling, thermal recycling, and chemical recycling. Material recycling is the method for reuse as material without chemical change. Owing to deterioration in quality, this method is not ideal for recycling. Thermal recycling is the method to recover energy by incinerating waste plastics. This method is easily applicable for a mixture of waste plastics. Carbon resource could not be recycled, although this method is extensively used.

However, chemical recycling of waste plastics has attracted great attention from the environmental viewpoint in recent year. Chemical recycling is classified into thermal decomposition, depolymerization (monomerization), and partial oxidation (gasification).

Although several technologies have already been commercialized, they have problems such as apparatus corrosion caused by strong acid catalyst and slow reaction rate. Recently, sub- and supercritical water ($T_c = 647.3$ K, $P_c = 22.1$ MPa) have been applied to recycle waste plastics. At near critical point, water density can be changed continuously from liquid state to gas state by controlling temperature and pressure. Furthermore, dielectric constant of sub- and supercritical water is 2–30, whereas, ambient water has about 80. Therefore, water is a good solvent of organic compound at high temperature and pressure. At the same time, ion product presenting hydrolysis power has maximum value at 473–573 K and accelerates ionic reaction. On the other hand, ion product becomes small in supercritical region. Large value in ion product indicates that hydrogen ion and hydroxyl ion concentrations are high.

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0141-3910/$ - see front matter © 2006 Elsevier Ltd. All rights reserved.
doi:10.1016/j.polymdegradstab.2006.02.009

Therefore, ionic reaction is dominant and hydrolysis may be accelerated because water behaves as both acid and alkali catalysts. Therefore, sub- and supercritical water is paid much attention as media for waste plastics recycling [1–3].

Polycondensation polymers are easily monomerized by depolymerization reaction such as hydrolysis. Polycondensation polymer containing ether bond, ester bond, or acid amide bond has been easily monomerized by solvolysis (hydrolysis and alcoholysis) in supercritical water and supercritical methanol. Polyethylene terephthalate (PET), which is one of polycondensation polymers, has been studied for developing new chemical recycling technologies in terms of hydrolysis [4] or solvolysis [5–7].

Nylon 6 containing acid amide bond has been produced by polymerization with ring-opening reaction of ε-caprolactam. It has been used extensively for clothing, fiber and packaging. The thermal degradation of nylon 6 has been studied for over 60 years [8]. It has been reported [9–13] that the pyrolysis product of nylon 6 is ε-caprolactam and this process has two degradation processes; an intramolecular back-biting process and a hydrogen transfer reaction leading to scission of the C–N bond β to the amide group. For example in recent research, Bockhorn et al. [14] studied decomposition of nylon 6 at 653–733 K and 1 MPa. When reaction condition was over 683 K and 100 min, nylon 6 was decomposed 100% and the yield of ε-caprolactam was over 92%. As temperature increased, the yield of by-products was increased. Additionally, the yield of ε-caprolactam was increased by reduction of by-product when catalyst was added.

Recently, the hydrolysis of nylon 6 in sub- and supercritical water has been paid much attention. This process is expected to produce ε-aminocaproic acid and ε-caprolactam without producing by-product. Sato and Ikushima [15] studied the decomposition of nylon 6 using a batch-type reactor at 573–653 K. ε-caprolactam was recovered 86% and ε-aminocaproic acid was recovered 5.5% at 653 K, 20 MPa and 10 min. When they used a flow-type reactor, the yield of ε-caprolactam was about 90% at 658 K, 30 MPa and 6–7 min.

In our previous work [16], we reported depolymerization of nylon 6 in sub- and supercritical water at 573–673 K. In the previous study, the maximum yield of ε-caprolactam was 60% and that of ε-aminocaproic acid was 40%. In this study, we aim to increase the yield of ε-caprolactam. The influences of reaction temperature, time and water density were investigated and the reaction scheme which is necessary to industrialize the process is also proposed.

2. Experimental

2.1. Experimental method

The experimental apparatus (AKICO Co., Japan) is shown in Fig. 1. The apparatus consists of a batch-type reactor (about 5 cm^3 inner volume) made of Hastelloy and heating electric furnace. Nylon 6 was purchased from Sigma–Aldrich Co., ε-aminocaproic acid with purity more than 97% and ε-caprolactam with purity more than 98% were from Wako Pure Chemical Industries, Ltd.

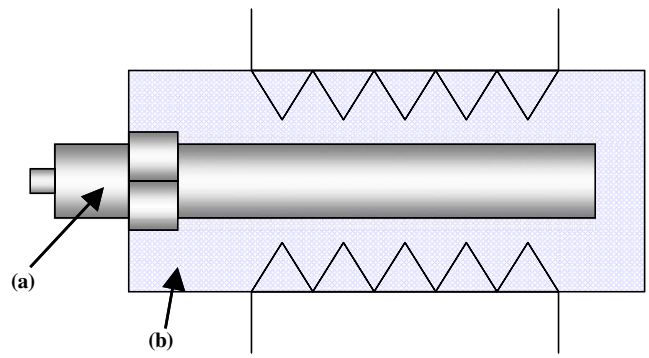

Fig. 1. The apparatus (a) reactor and (b) electric furnace.

The reactions were carried out at temperatures between 573 and 673 K under the estimated pressure of up to 35 MPa (water amount charged in a reactor 0.83–3.79 g/5 cm^3 – reactor) for the reaction time of 5–60 min. In this study, water amount was changed to change the pressure from 20 to 35 MPa. About 0.1 g of nylon 6 and degassed water were charged into a batch-type reactor. The remaining air in the reactor was purged with argon gas. The reactor was set into an electric furnace which is previously heated to the desired temperature. After a certain time, the reactor was quenched quickly by immersing water and the products were collected into test tubes.

2.2. Analysis method

The reaction product was separated into solid and liquid phases. The solid product was little. Additionally, there was no solid phase produced or remained for runs with sufficient reaction time. The liquid product was analyzed with a high performance liquid chromatograph (HPLC). The liquid product was diluted with water to 0.2 wt% for the analysis. The column was Shodex Asahipak® ODP of 150 mm × 4.6 mmϕ. The column was kept at 303 K and the mobile phase was the mixture of 0.1% acetic acid (Wako Pure Chemical Industries, Ltd.)/acetonitrile = 90/10 (Wako Pure Chemical Industries, Ltd.) with the flow rate of 0.5 ml/min. The detector was UV–VIS detector (Jasco UV-970) to analyze the components' quantity.

3. Results and discussion

3.1. Product

Fig. 2 shows the temperature history within the reactor during a run. According to this figure, it took about 3 min to reach the desired temperature. Therefore, the starting time for a reaction was defined to be 0 after 3 min of the reactor loading in the furnace.

The main products of hydrolysis of nylon 6 in sub- and supercritical water are ε-caprolactam and ε-aminocaproic acid. Time course of product appearance is summarized in

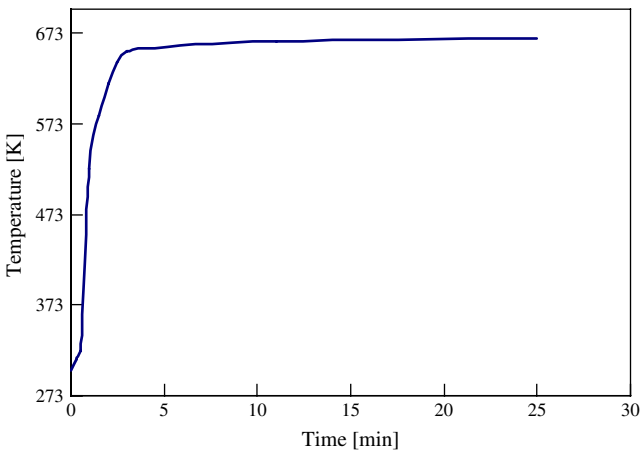

Fig. 2. Temperature change in the reactor as a function of time (set temperature: 673 K).

Table 1. From this table, the product was slurry at 573 K, 10 min. Since the monomer can be dissolved in water, we considered that the solid phase was nylon 6 polymer or oligomer. As the reaction time increased (at 573 K, 30 min), the reaction product was only a clear liquid. At lower temperature (573–603 K), product was only a transparent colorless liquid at 30 min. At 633 K and 673 K, product was only a transparent colorless liquid at 10 min and 5 min, respectively. Therefore, the higher is the reaction temperature, the shorter is a time when the reaction product was only a transparent colorless liquid. This result shows that nylon 6 was decomposed completely to monomer or small molecules dissolved in water. These results suggested that nylon 6 was decomposed into low molecule by sub-and supercritical water in a short time. We focused on the yield of ε-caprolactam and ε-aminocaproic acid as the main products of nylon 6 decomposition because they are monomers for the polymerization of nylon 6.

The yield of monomer products (ε-caprolactam and ε-aminocaproic acid) was defined as:

$$\text{Yield } [\%] = \text{ moles of products/moles of nylon 6 units} \times 100 \quad (1)$$

Table 1
Reaction product appearance

T (K)	P (MPa)	t (min)	Product
573	20–35	5–10	Slurry
		30–60	A transparent colorless liquid
603	20–35	5–10	Slurry
		30–60	A transparent colorless liquid
633	20–35	5	Slurry
		10–60	A transparent colorless liquid
673	25–35	5–60	A transparent colorless liquid

3.2. Yield of ε-aminocaproic acid

Fig. 3 shows time course of the yield of ε-aminocaproic acid at 573 K, 603 K, 633 K and 673 K. Each symbol represents reaction pressure. However, water densities are changed with temperature and pressure change. Therefore, water densities were described in caption. The maximum yield of ε-aminocaproic acid reached only about 13% at 633 K. At all reaction conditions the yield of ε-aminocaproic acid increased and had a maximum value. Then, ε-aminocaproic acid yield rapidly decreased. These results suggested that ε-aminocaproic acid is a intermediate product of hydrolysis of nylon 6 in sub- and supercritical water.

First, we discuss the influence of reaction temperature and time. At 573 K, 603 K, and 633 K, the yield had the maximum value at reaction time 30, 10, and 5 min, respectively. Therefore, the maximum yield of ε-aminocaproic acid was observed at shorter reaction time and at higher reaction temperature. In supercritical region at 673 K, only small amount of ε-aminocaproic acid was produced, because ε-aminocaproic acid was further converted to the other molecules rapidly. Thus, temperature has a considerable influence on ε-aminocaproic acid production.

Next, we discuss the influence of reaction water density. The water amount charged in the reactor was varied from 0.83 to 3.79 g/5 cm^3 – reactor. In subcritical region, liquid phase and vapor phase coexist in the reactor. The reaction may be occurred in liquid phase in subcritical region and the density of liquid phase varies slightly depending on the temperature. The estimated density of liquid phase for temperatures of 573–633 K is shown in Fig. 3. On the other hand, there is one homogeneous phase in supercritical region. The density of the fluid depends directly on the charged amount of water at 673 K and shown in Fig. 3. At subcritical temperature, there is no obvious influence of the density on the yield of ε-aminocaproic acid. It is reported that hydrolysis is affected by water density in supercritical region [17]. Therefore, the influence of water density on hydrolysis of nylon 6 to ε-aminocaproic acid was expected at supercritical temperature. However, the effect of water density was not obviously observed in case of the yield of ε-aminocaproic acid at 673 K because the produced amount was too small.

3.3. Yield of ε-caprolactam

The yield of ε-aminocaproic acid was considerably smaller than that of ε-caprolactam at all reaction conditions. Since ε-caprolactam is a starting monomer to produce nylon 6 by condensed polymerization, higher yield of ε-caprolactam is preferred.

Fig. 4 shows time course of the yield of ε-caprolactam at 573 K, 603 K, 633 K and 673 K. Each symbol represents reaction pressure. However, water densities are changed with temperature and pressure change. Therefore, water densities were described in caption. The highest yield of ε-caprolactam reached about 85% at 573 K, 60 min. The yields had maximum values for temperatures of 573 K, 603 K, and 633 K at

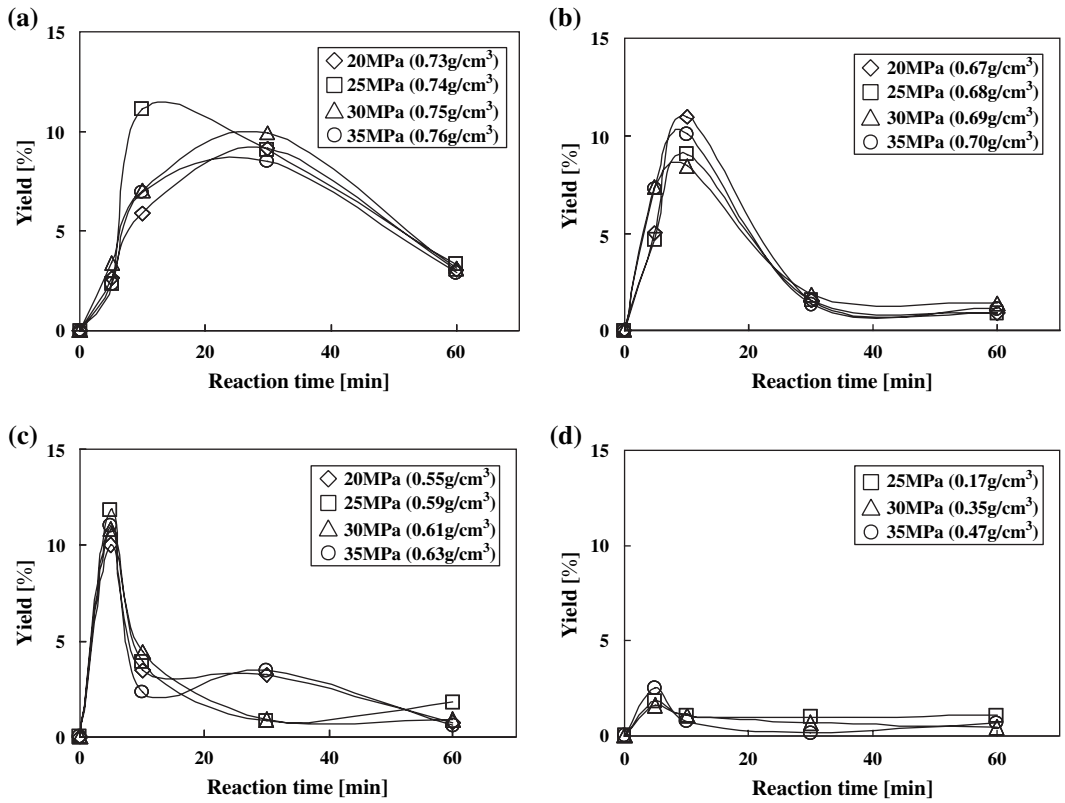

Fig. 3. Time course of the yield of ε-aminocaproic acid at (a) 573 K, (b) 603 K, (c) 633 K and (d) 673 K.

reaction time of 60, 30, and 10 min, respectively. The higher is the reaction temperature, the faster is the ε-caprolactam production rate. At temperatures above 633 K, the yield of ε-caprolactam once had a maximum value and then decreased with reaction time. This indicates that secondary reaction occurred in longer reaction time where ε-caprolactam degraded. When the temperature was lower than 603 K, ε-caprolactam produced was stable.

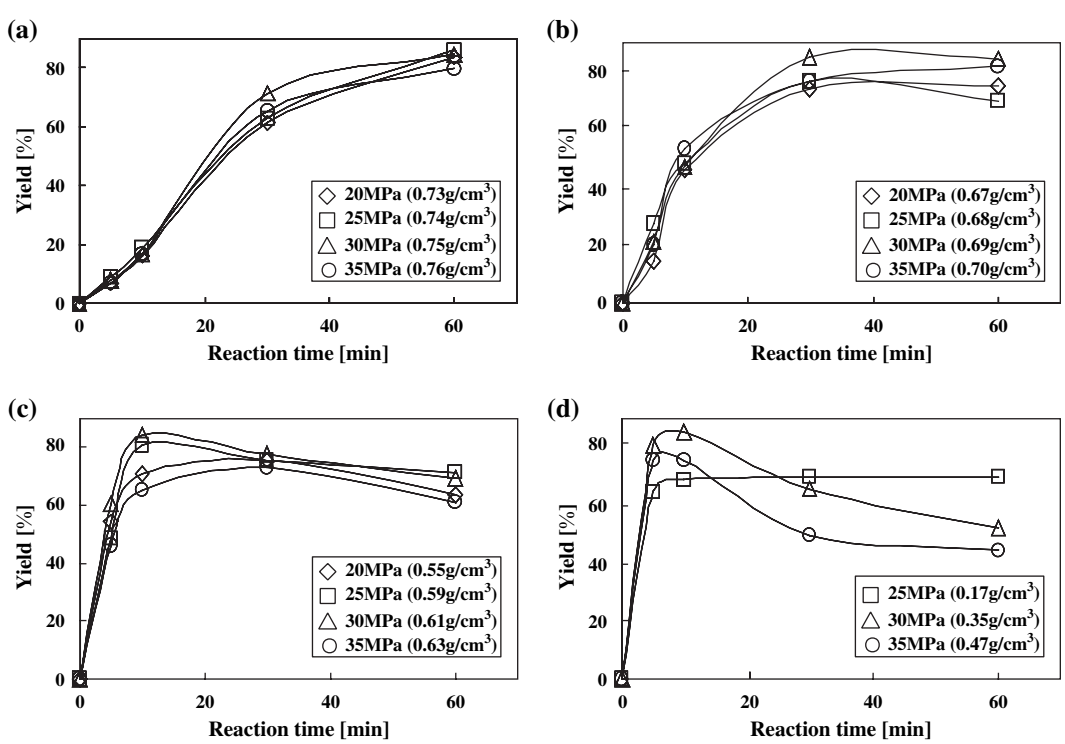

Fig. 4. Time course of the yield of ε-caprolactam at (a) 573 K, (b) 603 K, (c) 633 K and (d) 673 K.

Next, we discuss the influence of amount of water charged in a reactor or water density. At temperatures below 633 K, there is no apparent influence of water density, because the liquid phase density does not change largely. However, at 673 K, the ε-caprolactam production rate is increased as water density increased because of supercritical region. In addition, secondary reaction tended to occur when water density increased. In supercritical region, density change may influence reaction environment. At higher density, the dissolution power is higher where both dissolution of molecules into supercritical fluid phase and penetration/dissolution of water into polymer phase increase. Accordingly, it is expected that nylon 6 is dissolved easily and reaction is progressed in homogeneous phase. These results are summarized as follows: decomposition of ε-aminocaproic acid to ε-caprolactam was influenced by reaction temperature at every condition and by water density at supercritical condition.

3.4. Reaction scheme

Reaction products were not produced except ε-aminocaproic acid and ε-caprolactam at reaction initial stage. As shown in Fig. 4, the yield of ε-caprolactam increased in s-shape with reaction time at 573 K and 603 K. This kind of time course on the product yield can be seen on the typical consecutive reaction. Additionally, as shown in Fig. 3, the yield of ε-caprolactam increased with decreasing the yield of ε-aminocaproic acid. This result suggested that nylon 6 decomposition proceeded as a consecutive reaction. Based on our experimental results, the reaction scheme of nylon 6 decomposition is represented by Fig. 5. Nylon 6 was decomposed by hydrolysis into ε-aminocaproic acid followed by cyclodehydration to ε-caprolactam or decomposition further to lower molecules. Additionally, ε-caprolactam is decomposed to secondary products in higher temperature and longer reaction time. It is reported [13] that thermal decomposition of nylon 6 has two reaction processes; an intramolecular back-biting process and a hydrogen transfer reaction leading to scission of the C–N bond β to the amide group. On the other hand, hydrolysis of nylon 6 using sub- and supercritical water may be a single consecutive reaction process. Therefore, reaction control is easy and the monomer can be easily collected by sub- and supercritical method.

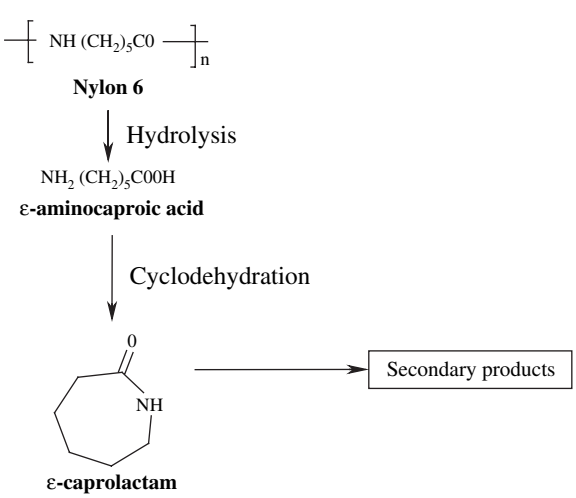

Fig. 5. Reaction scheme of nylon 6 decomposition in hydrothermal and supercritical water.

3.5. Reaction kinetics

On the basis of our results, a simple kinetic model was developed and used to analyze the data. We assumed that the reaction is a consecutive reaction and the decomposition of nylon 6 is described by the following reaction equation.

$$\text{Nylon 6} \xrightarrow{k_1} \text{ε-aminocaproic acid} \xrightarrow{k_2} \text{ε-caprolactam} \tag{2}$$

The production rates of ε-aminocaproic acid and ε-caprolactam are given by Eqs. (3) and (4).

$$\frac{dC_{\text{ε-aminocaproic acid}}}{dt} = k_1 C_{\text{nylon 6}} - k_2 C_{\text{ε-aminocaproic acid}} \tag{3}$$

$$\frac{dC_{\text{ε-caprolactam}}}{dt} = k_2 C_{\text{ε-aminocaproic acid}} \tag{4}$$

Eqs. (3) and (4) are integrated with the initial conditions; $C_{\text{nylon 6}} = C_{\text{nylon 6, 0}}$, $C_{\text{ε-aminocaproic acid}} = C_{\text{ε-caprolactam}} = 0$ at $t = 0$. Then, the concentrations of monomers, $C_{\text{ε-aminocaproic acid}}$ and $C_{\text{ε-caprolactam}}$, are expressed by Eqs. (5) and (6).

$$C_{\text{ε-aminocaproic acid}} = C_{\text{nylon 6, 0}} \frac{k_1}{k_1 - k_2}\left(e^{-k_2 t} - e^{-k_1 t}\right) \tag{5}$$

$$C_{\text{ε-caprolactam}} = C_{\text{nylon 6, 0}}\left(1 - e^{-k_1 t}\right) - C_{\text{ε-aminocaproic acid}} \tag{6}$$

The values of k_1 and k_2 were determined by the comparison of observed time variation of $C_{\text{nylon 6}}$, $C_{\text{ε-aminocaproic acid}}$ and $C_{\text{ε-caprolactam}}$. As an example, comparison between experimental results and model prediction at 573 K, 25 MPa was depicted in Fig. 6. The values determined are $k_1 = 5.5 \times 10^{-4}\ s^{-1}$ and $k_2 = 1.0 \times 10^{-2}\ s^{-1}$ for a curve fitting with the least squares method.

At each reaction conditions, the k_1 and k_2 values were calculated in the same manner and listed in Table 2. The k_1 values were determined to be 5.0×10^{-4}–$4.5 \times 10^{-3}\ s^{-1}$. The k_2 values were determined to be 1.0×10^{-2}–$1.8 \times 10^{-2}\ s^{-1}$. For every reaction condition, the k_2 was larger than the k_1. These results suggested that hydrolysis of nylon 6 to ε-aminocaproic acid was slower reaction and cyclodehydration of ε-aminocaproic acid to ε-caprolactam was faster reaction. With increased reaction temperature, the k_1 and k_2 values were increased. Additionally, the k_1 value was more temperature sensitive than the k_2 value. Thus, hydrolysis of nylon 6 to ε-aminocaproic acid is more easily controlled by controlled temperature than cyclodehydration of ε-aminocaproic acid to ε-caprolactam.

3.6. Activation energy

Fig. 7 shows the Arrhenius plot of the k_1 and k_2 values obtained in the range of 573–673 K. The activation energy

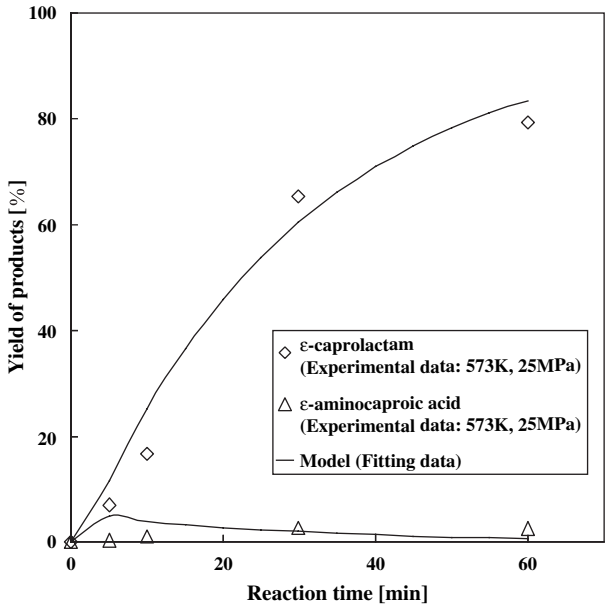

Fig. 6. Comparison between experimental results and model prediction ($k_1 = 5.5 \times 10^{-4}$, $k_2 = 1.0 \times 10^{-2}$).

(E_a) was calculated from the slope and the respective Arrhenius constant (A) was calculated from the intercept. Hydrolysis activation energy of nylon 6 to ε-aminocaproic acid was found to be $E_{a1} = 97.1$ kJ mol^{-1} at 573–633 K and $E_{a1} = 5.4$ kJ mol^{-1} at 633–673 K. Cyclodehydration activation energy of ε-aminocaproic acid to ε-caprolactam was found to be $E_{a2} = 16.1$ kJ mol^{-1}. In supercritical region, hydrolysis of nylon 6 to ε-aminocaproic acid is more accelerated than cyclodehydration of ε-aminocaproic acid to ε-caprolactam. It is reported [8] that thermal decomposition activation energy of nylon 6 was 40–220 kJ mol^{-1}. These results suggested that decomposition of nylon 6 to ε-caprolactam in sub- and supercritical water is more feasible method compared with thermal decomposition of nylon 6.

Table 2
The first-order rate constants (k_1 and k_2) determined

Temperature (K)	Pressure (MPa)	Water amount (g)	Water density (g/cm^3 –reactor)	k_1 (s^{-1})	k_2 (s^{-1})
573	20	3.67	0.73	5.0×10^{-4}	1.0×10^{-2}
	25	3.72	0.74	5.5×10^{-4}	1.0×10^{-2}
	30	3.75	0.75	6.5×10^{-4}	1.1×10^{-2}
	35	3.79	0.76	5.7×10^{-4}	1.1×10^{-2}
603	20	3.33	0.67	1.1×10^{-3}	1.5×10^{-2}
	25	3.40	0.68	1.2×10^{-3}	1.6×10^{-2}
	30	3.46	0.69	1.3×10^{-3}	1.6×10^{-2}
	35	3.51	0.70	1.3×10^{-3}	1.6×10^{-2}
633	20	2.74	0.55	3.6×10^{-3}	1.6×10^{-2}
	25	2.94	0.59	3.7×10^{-3}	1.6×10^{-2}
	30	3.07	0.61	4.0×10^{-3}	1.7×10^{-2}
	35	3.16	0.63	4.5×10^{-3}	1.7×10^{-2}
673	25	0.83	0.17	4.0×10^{-3}	1.8×10^{-2}
	30	1.77	0.35	4.4×10^{-3}	1.8×10^{-2}
	35	2.37	0.47	4.2×10^{-3}	1.8×10^{-2}

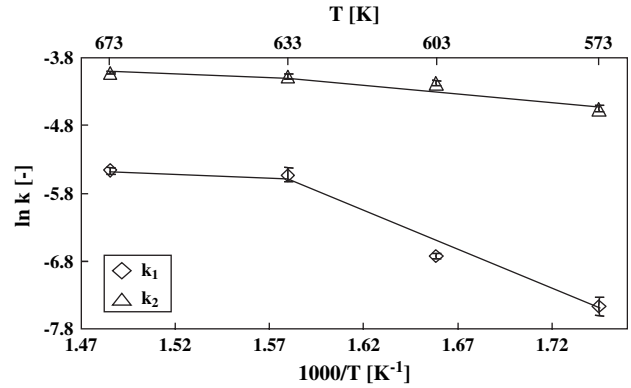

Fig. 7. Arrhenius plot of the rate constant for decomposition of nylon 6.

4. Conclusion

Nylon 6 was decomposed into low molecule in sub- and supercritical water in a short time. The highest yield of ε-caprolactam reached 85%. The higher is the reaction temperature, the faster is the ε-caprolactam production rate. At temperatures above 633 K, the yield of ε-caprolactam decreased with reaction time due to secondary reaction in long reaction time. At supercritical temperature, the ε-caprolactam production rate was increased with water density. Secondary reaction was promoted with water density. The yield of ε-aminocaproic acid reached only about 13%. The reaction from nylon 6 to ε-aminocaproic acid and then to ε-caprolactam proceeded as a consecutive reaction where ε-aminocaproic acid was the intermediate product. Nylon 6 was decomposed by hydrolysis into ε-aminocaproic acid followed by cyclodehydration to ε-caprolactam or decomposition further to lower molecules. The reaction rate constant, k_1 and k_2 were determined and analyzed to evaluate activation energy. These results suggested that hydrolysis of nylon 6 to ε-aminocaproic acid is more accelerated than cyclodehydration of ε-aminocaproic acid to ε-caprolactam in supercritical region.

Acknowledgement

The financial support of Kumamoto University 21st Century Center of Excellence (COE) Program ‘‘Pulse Power Science’’ is gratefully acknowledged.

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